

Arjun Dadhwal

PhD Student, Computer Science | Scientific ML, Visualization, TDA & Super-Resolution
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EDUCATION

Tulane University

PhD Student, Computer Science

New Orleans, LA

August 2025 - Present

- GPA: 3.89/4.00; Research Assistant advised by Prof. Brian Summa.
- Research areas: scientific visualization, topological data analysis, and topology-aware super-resolution for scientific scalar fields.

Arizona State University

M.S. Computer Science, Artificial Intelligence Specialization

Tempe, AZ

May 2025

- GPA: 3.90/4.00; Full Tuition Waiver Award; coursework included Statistical Machine Learning, Data Visualization, Data Mining, and Data Processing at Scale.

Arizona State University

B.S. Computer Science

Tempe, AZ

May 2024

- GPA: 3.99/4.00; Summa Cum Laude; New American University Scholarship; Dean's List.

PUBLICATIONS & PRESENTATIONS

Topology-Inspired Fine-Tuning for Wind-Field Super-Resolution

Manuscript completed; arXiv preprint in preparation. **Poster forthcoming** at TopolnVis 2026 (IEEE VIS 2026)

RESEARCH EXPERIENCE

Tulane University

Research Assistant

New Orleans, LA

August 2025 - Present

- Develop topology-aware methods for wind-field super-resolution using persistence diagrams, merge trees, physical metrics, and topology-inspired training objectives.
- Built reproducible evaluation pipelines spanning pointwise, physical, distributional, and topological metrics, including cross-implementation validation of persistence and merge-tree analyses.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, Swift, Dart, C#

ML/Data: PyTorch, TensorFlow, NumPy, pandas, scikit-learn, SciPy, Matplotlib, XGBoost

Research: Scientific visualization, topological data analysis, scalar-field topology, persistence diagrams, merge trees, super-resolution, model evaluation

Systems/Tools: Git, Docker, Linux, PostgreSQL, MySQL, MongoDB, Spark/Hadoop, REST APIs

SELECTED PROJECTS

Topology-Inspired Wind-Field Super-Resolution

Python, TensorFlow/PyTorch, TTK/VTK, GUDHI, TDA

2026

- Evaluated neural wind-field super-resolution using PSNR/SSIM, physical and distributional measures, persistence diagrams, and merge trees across a 168-field benchmark.
- Developed topology-inspired fine-tuning with gradient, critical-value, and persistence supervision; improved bottleneck and Wasserstein persistence-diagram agreement while retaining strong conventional fidelity.
- Cross-validated persistence and merge-tree computations across independent implementations and constructions.

Stable Segmentation Inference Service

Python, FastAPI, CLIPSeg/SAM2, Docker, Testing

2026

- Built a prototype versioned segmentation service with mock, CLIPSeg, and SAM2 backends; implemented CLI/web clients, capability endpoints, and API compatibility checks.
- Expanded automated test coverage to 338 passing tests across API behavior, client workflows, request validation, and backend contracts.

Rich Text RAG for arXiv Papers

Python, PyMuPDF, feedparser, FAISS/RocksDB

2024

- Created a robust arXiv ingestion pipeline with paginated Atom queries, rate limiting, retries, PDF/source URL construction, deduplication, and NDJSON export.
- Stream-downloaded PDFs with checksum checks, extracted page-level text, normalized metadata, and produced JSON records for downstream vector/database ingestion.

EARLIER EXPERIENCE

Arizona State University

Teaching Roles (UGTA, TA, IA)

Tempe, AZ

January 2024 - May 2025

- Led office hours and exam reviews, graded assignments, produced tutorial materials, and supported course delivery for undergraduate and graduate CS courses.

Sandler Training

Software Engineer (Capstone)

Winnipeg, MB

May 2023 - December 2023

- Led development of a full-stack, cross-platform mobile habit-tracking application using React Native and Firebase services for authentication and real-time data.

CBTB Lab, Indian Institute of Technology Roorkee

Software Engineer Intern

India

June 2022 - August 2022

- Migrated Dhadkan, a Google Play Store remote health-monitoring app, from React Native to Flutter to improve app stability and responsiveness.